

Developer Azure CLI access pattern (PIM, dev subscription)

Developer temporary Azure CLI access

Track 2 (identity) — give **developers** time-bound `az` capability on the **dev subscription** without shared admin accounts or permanent Contributor.

Problem

Developers need Azure CLI for:

- Inspecting dev VMs, NSGs, and private endpoints
- `az acr login` + `local podman push` to dev registry
- Reading dev Key Vault secrets for local testing

VMs (runner/runtime) use **UAMI** (`az login --identity`). **Humans** on laptops or Bastion dev hosts use **Entra ID + temporary PIM activation**.

Banking default workflow

```
sequenceDiagram
    participant Dev as Developer
    participant PIM as Entra PIM
    participant Az as Azure CLI
    participant Azure as Dev subscription

    Dev->>PIM: Activate eligible role (≤8h, MFA)
    PIM-->>Dev: Time-bound RBAC active
    Dev->>Az: az-developer-login.sh
    Az->>Azure: API calls with user token
    Note over Dev,Azure: Token expires when PIM activation ends
    Dev->>Az: az logout
```

Step	Action
1	Patrick adds developer to <code>grp-platform-developers</code> (Entra group)
2	Platform applies PIM eligible roles via <code>developer-cli-access</code> Terraform module (dev only)
3	Developer activates role in Entra → PIM → My roles (justify, MFA, ≤8h)
4	Developer runs <code>./scripts/az-developer-login.sh</code>
5	Developer uses <code>az</code> until activation expires; <code>az logout</code> when done

PIM eligible roles (dev scope)

Role	Scope	CLI examples
Reader	Dev subscription or RG	<code>az vm list</code> , <code>az network nsg rule list</code>
AcrPush	Dev ACR	<code>az acr login --name <acr>; podman push</code>
Key Vault Secrets User	Dev KV	<code>az keyvault secret show</code>

Never on prod/UAT subscriptions for developer group. **Never** Contributor or Owner for routine dev CLI.

Where to run

Environment	Login method
Developer laptop (macOS/WSL)	<code>scripts/az-developer-login.sh</code> after PIM
Dev Linux VM via Bastion	<code>/opt/compliance/bootstrap/az-login-developer.sh</code> (same flow)
GitHub runner VM	UAMI only — not developer Entra login
UAT/prod runtime VM	Runtime UAMI only — pull-only

IaC

```
module "developer_cli" {
  source = "../../modules/developer-cli-access"

  developer_group_object_id = var.entra_group_platform_developers_id
  dev_subscription_id       = var.dev_subscription_id
  dev_resource_group_id    = azurerm_resource_group.dev.id
  acr_id                   = module.acr.acr_id
  key_vault_id             = azurerm_key_vault.dev.id
}
```

Configure **PIM activation max duration** (8h) in Entra role settings to match `pim_activation_max_hours`.

Anti-patterns

Don't	Do instead
Shared <code>deploy</code> user with password	Individual Entra user + PIM
Service principal secret on laptop <code>.env</code>	PIM + <code>az login</code>
Permanent Reader on dev sub	PIM eligible only
<code>az login</code> with prod subscription for devs	<code>AZURE_DEV_SUBSCRIPTION_ID</code> in script
Developer <code>az</code> on UAT/prod runtime VM	UAMI automation only

Related

- — developer CLI + consultant discovery patterns
- §6.2 — UAMI on VMs
- §2 Identity

Industry references

Full map:

Topic	Source
Least privilege / JIT	Entra PIM NIST SP 800-207
Managed identity (VMs)	Managed identities overview
Azure CLI identity login	Sign in with Azure CLI — managed identity
Banking access control	FFIEC IS Handbook

Document history

Version	Date	Notes
0.1	2026-05-28	Developer PIM + az login workflow